CONDITION BASED ASSESSMENT
QUICK SCANNING REPORT
SBUM DIAGNOSTIC JOHOR
TNB BAHAGIAN PEMBAHAGIAN JOHOR

FUNCTIONAL LOCATION (ERMS)	:	JJBU/PCE/H01031
FUNCTIONAL LOCATION (SITE)	:	JJBU/PCE/H01031
SUBSTATION NAME (ERMS)	:	TAMAN ULAM RAJA
SUBSTATION NAME (SITE)	:	TAMAN ULAM RAJA JOHOR BAHRU
COORDINATE	:	1.579336, 103.718377
SUBSTATION TYPE	:	PCE
AREA	:	JOHOR BAHRU
STATE	:	JOHOR
PO	:	42234207
DATE	:	14 JUL 2026

SCANNED BY		
THERMOGRAPHY	:	MUHAMMAD OMAR RIDZUAN (2013MY20N010)

EQUIPMENT/TOOLS		
THERMOGRAPHY	:	FLIR T640 (SN 55912229) (CALIBRATION DUE DATE: 26 DEC 2026)

PREPARED BY		
NAME OF CONTRACTOR	:	**EET ENGINEERING SERVICES SDN BHD**
ADDRESS	:	NO.38, JALAN SS19/1D,
	:	47500, SUBANG JAYA, SELANGOR.

SUBSTATION SIGNBOARD



VISUAL DEFECT SUMMARY

NO.	EQUIPMENT	DEFECT DESCRIPTION	ELECTRICAL	NON-ELECTRICAL
1	SUBSTATION	NO PADLOCK – SWG ROOM		X
2	RUBBER MAT	NO RUBBER MAT		x

SUBSTATION CONDITION

SUBSTATION OVERVIEW



SIGNBOARD



SWITCHGEAR



SWITCHGEAR NAMEPLATE



TRANSFORMER 1

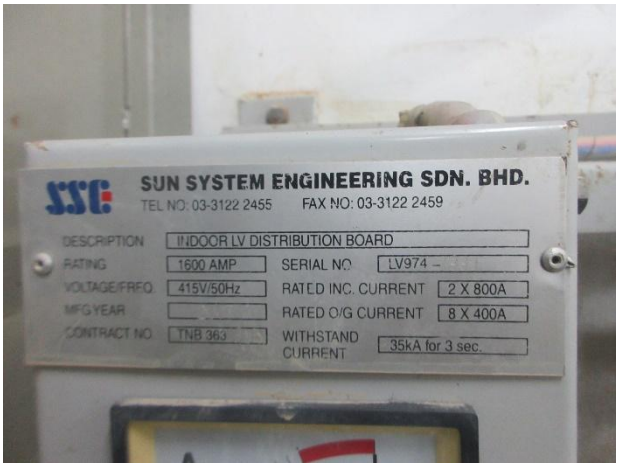


TRANSFORMER 1 NAMEPLATE



SUBSTATION CONDITION

TRANSFORMER 2	TRANSFORMER 2 NAMEPLATE
N/A	N/A

LVDB TX1	LVDB TX1 NAMEPLATE
	

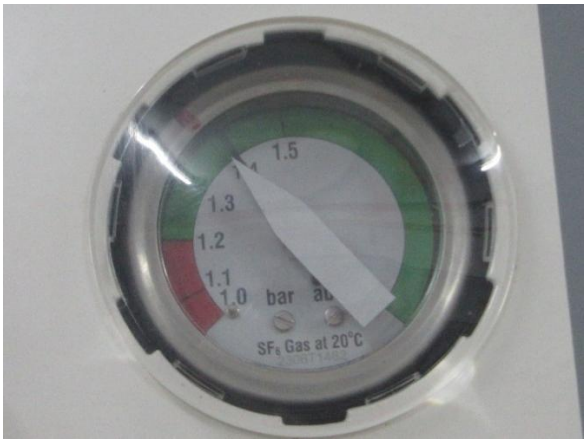
FEEDER PILLAR TX2	FEEDER PILLAR TX2 NAMEPLATE
N/A	N/A

SUBSTATION CONDITION

EFI



SF6 INDICATOR



BATTERY 1



BATTERY 1 NAMEPLATE



BATTERY 2



BATTERY 2 NAMEPLATE



SUBSTATION CONDITION

FIRE EXTINGUISHER	FIRE EXTINGUISHER NAMEPLATE
	

SWITCHGEAR PANEL NAME		LAST PREVENTIVE MAINTENANCE
		

VISUAL DEFECT

RUBBER MAT

NO RUBBER MAT

SUBSTATION

NO PADLOCK – SWG ROOM

NORMAL/DEFECT STICKER

NORMAL STICKER


CBM CONDITION	79
	NORMAL

DISTRIBUTION SUBSTATION (PCE) CBA TEST SHEET

STATE / AREA:	JOHOR / JOHOR BAHRU	**DATE**	14/07/26			**FUNCTIONAL LOCATION:**	
PE NAME:	TAMAN ULAM RAJA	**TIME IN / OUT**	1900	/	1930	JJBU/PCE/H01031	
V.I REF. NO	79	**TEV BACK GROUND / HUMIDITY**	5 dB	/	64 %	**BACKGROUND TEMP : 32.3°C**	

SWITCHGEAR

DETAIL								TEMPERATURE							ULTRASOUND			TEV			REMARK
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLU)	COMP	Tmin (°C)	Tmax (°C)	ΔT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE/ CYCLE	TYPE CHAR.	
1	JJB96240	SK KG MAJU JAYA	CLOSE		XLPE			CABLE	33.2	33.2	0.0	33.2	209		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
2	JJB96241	TX 500 KVA	CLOSE		XLPE			CABLE	33.1	33.1	0.0	33.1	210		0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
3	JJB96242	PPU SETIA TROPIKA CB 06	CLOSE		XLPE			CABLE	33.4	33.4	0.0	33.4	211		0			6	0		
								BREAKER													
								TOP PANEL													
								PT													
4								CABLE							0			5	0		
								BREAKER													
								TOP PANEL													
								PT													
OVERVIEW								CABLE				33.2	204								
								BREAKER													
								TOP PANEL				33.2	208								
								PT													

LOCAL TRANSFORMER

DETAILS					THERMOGRAHY					ULTRASOUND		DETAILS				THERMOGRAHY						ULTRASOUND		
TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	ΔT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/SIZE	COMP.	Tmin(°C)	Tmax(°C)	ΔT	Avg (°C)	FILE	dB	FILE	TYPE CHAR.	REMARKS
TX1	EWT/91260	XLPE		HT CABLE	35.2	35.5	0.3	35.4	215	0		TX3			HT CABLE									
				HT BUSHING	37.5	37.9	0.4	37.7	214	0					HT BUSHING									
		XLPE		LV CABLE	35.4	35.9	0.5	35.7	217						LV CABLE									
				LV BUSHING	37.2	37.6	0.4	37.4	216						LV BUSHING									
				BODY	35.6	41.8	6.2	38.7	212						BODY									
TX2				HT CABLE								TX4			HT CABLE									
				HT BUSHING											HT BUSHING									
				LV CABLE											LV CABLE									
				LV BUSHING											LV BUSHING									
				BODY											BODY									

LV DISTRIBUTION BOARD/FEEDER PILLAR/OTHERS

1	LVDB / FP 1	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10									
	CABLE TYPE / SIZE	XLPE	XLPE						XLPE	XLPE		XLPE											
2	LVDB / FP 2	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10									
	CABLE TYPE / SIZE																						

NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	ΔT	FILE	Avg (°C)	REMARK
	INC 1							
	INC 2							
	F1							
	F2							
	F3							
	F4							
	F5							
	F6							
	F7							
	F8							
	Battery Bank 1		33.0	33.2	0.2	219	33.1	BC1: SUNPOWER M:SPS 30-10 L SN:11193-9638
	Battery Bank 2		34.0	34.2	0.2	220	34.1	BC2: BERKAT INSAF M:SEB100-30-10UL SN:B21219
	Black Box 2							
	Lightning Arrester							
	Dropout Fuse							
	Auto Recloser							
	LBS							

LVDB TX1	J-SLOTTED
Manufacturer :	SSE
Serial No. :	LV974-1999
Rating :	1600A
LVDB TX2	
Manufacturer :	
Serial No. :	
Rating :	

Tested by (Name & Signature) :

EET ENGINEERING SERVICES SDN BHD
(534556-W)
No. 38-2, 2nd Floor, Jalan SS 19/1D
47500 Subang Jaya, Selangor.
Tel: 603-5636 6201 Fax: 603-5636 7205

TNB Supervisor/Engineer (Name & Signature) :

NOR AZMI BIN ABDUL RAHMAN
(N/P 10073600)
Pen. Jurutek nik T/Kanan
Diagnostik (Substation - South)
Distribution Network Department, TNB

Ver. : 1-2019

Vi no mat, n p lock swg r

TYPE PCE 79

VISUAL INSPECTION CHECKLIST FOR 11kV SUBSTATION

TENAGA NASIONAL BERHAD (200866-W)

DATE:	14/07/26												
P/E Name (ERMS):	TAMAN ULAM RAJA							F/L NO (ERMS):	JJBU/PCE/H01031				
P/E Name (SITE):	TAMAN ULAM RAJA JOHOR BAHRU							F/L NO (SITE):	JJBU/PCE/H01031				
GPS Coordinate:	1.579336, 103.718377												
PCE Type:	INDOOR	☑		OUTDOOR		COMPACT		ATTACH BUILDING		PAT		OTHER	
PCE Area	NORMAL	☑		INDUSTRIAL		COASTAL							
Switcgear Type 1:	RMU OIL			RMU SF6		MRMU	☑	VCB		OCB		OTHER	
Manufacturer/Model:	EPE			CFC 2S+1F									
Switchgear Manufactured Date:	2024					Switchgear Rating :		12kV				RMU SF6 / OIL S/N :	23050047346
Switcgear Type 2:	RMU OIL			RMU SF6		MRMU		VCB		OCB			
Manufacturer/Modal:													
Switchgear Manufactured Date:						Switchgear Rating :						RMU SF6 / OIL S/N :	
No. of Transformer:	1												
Tx 1:	Type	H/S	KVA	500kVA		Construction Yr:	2024	Manufacturer:	EWT			S/N :	EWT/91260
Tx 2:	Type		KVA			Construction Yr:		Manufacturer:				S/N :	
Tx 3:	Type		KVA			Construction Yr:		Manufacturer:				S/N :	
Tx 4:	Type		KVA			Construction Yr:		Manufacturer:				S/N :	

KEJANGGALAN

1. Signboard:	Good	☑	Not Good		Remarks	
2. Earth connection:	Good	☑	Not Good		Remarks	
3. EFI Functioning:	Good		Not Good		Remarks	N/A
4. Heater Functioning:	Good		Not Good		Remarks	N/A
5. Labelling @Switchgear:	Good	☑	Not Good		Remarks	
6. Labelling @ LVDB:	Good	☑	Not Good		Remarks	
7. Gas Pressure:	Good	☑	Not Good		Remarks	
8. Tx Guard:	YES	☑	NO		Remarks	
9. Tx Oil Level (conservator tank):	Good		Not Good		Remarks	N/A
10. Tx Oil Leak:	YES		NO	☑	Remarks	
11. Tx Bushing:	Good	☑	Not Good		Remarks	
12. Top Demand Indicator (@LVDB):	Good	☑	Not Good		Remarks	
13. Link/fuse Holder:	Good	☑	Not Good		Remarks	
14. LV Termination:	Good	☑	Not Good		Remarks	
15. Rubber Mat:	YES		NO	☑	Remarks	
16. Trenching:	Good	☑	Not Good		Remarks	
17. Battery Voltage:	Good	☑	Not Good		Remarks	
18. Lighting:	Good	☑	Not Good		Remarks	
19. Roof Condition:	Good	☑	Not Good		Remarks	
20. Fire Fighting	Valid	☑	Expired		Remarks	Expiry Date: 11/10/2026 No. of missing portable : -
21. Fence/door condition:	Good	☑	Not Good		Remarks	
22. Cleanliness/Bushes:	YES		NO	☑	Remarks	
23. Vandalisme	YES		NO	☑	Remarks	
24. Other Findings:						

Tested by (Name & Signature) :

EET ENGINEERING SERVICES SDN BHD
(534556-W)
No. 38-2, 2nd Floor, Jalan SS 19/1D
47500 Subang Jaya, Selangor.
Tel: 603-5636 6200 Fax: 603-5636 7265

Date :

TNB Supervisor / Engineer (Name & Signature) :

NOR AZMI BIN ABDUL RAHMAN
(NP 10073600)
Sen. Juruteknik T/Kanan
Diagnostik (Substation - South)
Distribution Network Department, TNB

Date :

Ver. : 1-2019